ProTracker on the Technics KN7000

User manual: playing and editing Amiga modules on the KN7000's own panel, keybed and screen



Version of 27 October 2026, for the program in `kn5000_homebrew/demos/kn7000_protracker` (folder PROTRACK\ on the card). A port of pt2-clone by Olav Sørensen (8bitbubsy), the faithful C version of ProTracker 2.3D (BSD 3-Clause). The KN7000 side and this manual: GPLv3 or later.

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1. What this is

ProTracker is the Amiga's classic music tracker. A song (a *module*, a `.MOD` file) is a list of *positions*. Each position plays a *pattern* of 64 rows, and each row has four channels. A cell of a channel holds a note, the number of the sample it plays (1-31), and an effect: a command digit and a two-digit parameter, all in hexadecimal. The cursor moves through the cells, and the keyboard enters notes and digits. On the Amiga that keyboard is a computer keyboard and a mouse.

On the KN7000 the same program runs on the instrument. Its replayer plays the four channels through the instrument's effects processor (DSP), and its screen is drawn on the LCD. The KN7000 has no computer keyboard and no mouse, so **the panel and the keybed do their work**. The keybed plays and enters notes. The panel's buttons were chosen so that what each one does on the KN7000 suggests what it does in ProTracker: the sequencer's buttons are the song, the SOUND GROUP is a keypad of hex digits, PANEL MEMORY stores and recalls effects, and so on (section 4).

Every module the Amiga's ProTracker makes or plays with four channels and 31 samples works: ProTracker, NoiseTracker and Soundtracker modules. The examples in this manual use the music of *Pinball Fantasies*.

2. Starting and leaving

What goes on the SD card

The folder `PROTRACK\` at the card's root holds the program and your modules:

file	what
`KN7APP.BIN`, `PT.BIN`	the program
`MANIFEST.TXT`	the name the program menu shows: "ProTracker 2.3D"
`*.MOD`, `MOD.*`	the modules (`TABLE1.MOD`, `PTNEW01.MOD`, or Amiga-style names such as `MOD.JAZ`)

To add a module, copy it into `PROTRACK\`. The program lists every file there whose name ends in `.MOD` or starts with `MOD.`, in order of name, and steps through them in that order. A file with a long name is listed by the short 8.3 name the computer gave it on the card (`Space_Debris.mod` shows as `SPACE_~1.MOD`).

Starting

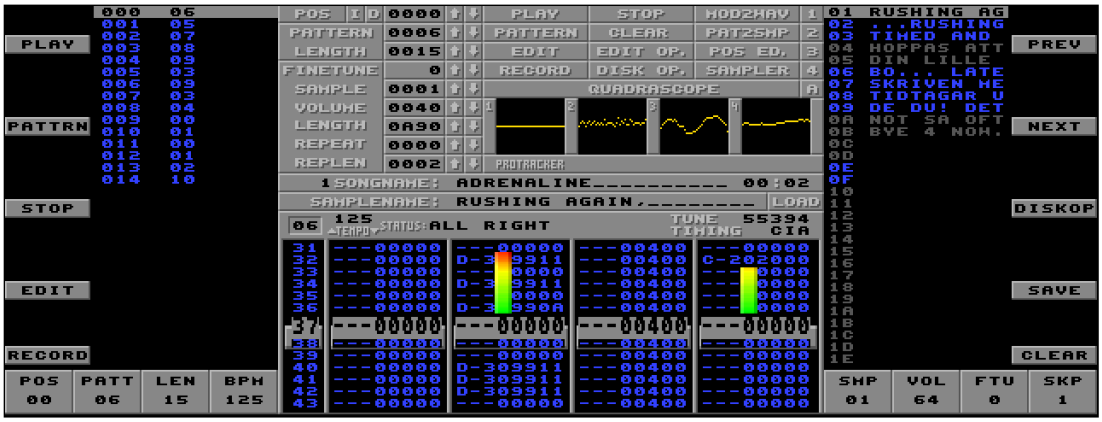
1. Insert the KN7LOAD floppy and the SD card.
2. Press `DISK MENU`, then `STYLE CONVERT` (left button 3), then `OK` on `KN7LOAD.CNV`.
3. In the program menu, select "ProTracker 2.3D" with the left buttons and press `RUN`.
4. The first module, by name, loads and plays. Its samples are sent to the DSP first: this takes between half a second and two seconds, and the status line says `SENDING SAMPLES...`

Leaving

Hold `EXIT` for 1.5 seconds. The instrument shows its message, then "Continue Next?": answer `NO` (right button 4) to go back to the instrument. A brief press of EXIT is ProTracker's Esc instead (section 11).

> **Save before leaving.** Edits live in the instrument's memory only until you save them (section 9). Leaving, or loading another module, loses unsaved edits (loading asks first).

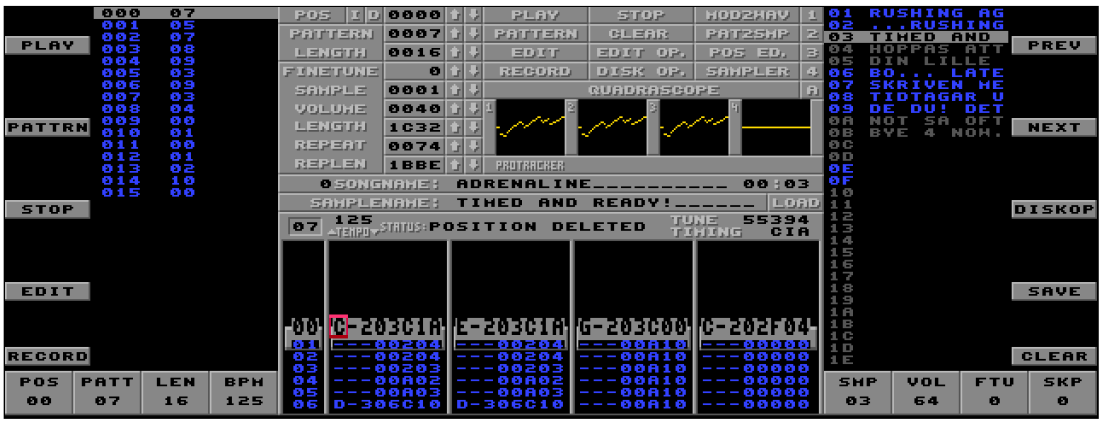
3. The screen



Playing: ProTracker's screen in the middle; the song's positions on the left, the samples on the right; the soft keys' tags at the outer edges; at the bottom, the values the MUTE buttons below change.

The LCD is 640 dots wide and ProTracker's screen 320, so ProTracker sits in the middle, as it looks on the Amiga (its pattern shows 13 rows instead of 15: the LCD is 240 lines high, the Amiga's screen 255). The 160 dots either side show what ProTracker hides on the Amiga behind other screens:

- **The soft keys' tags.** Beside each of the five buttons left and right of the LCD, the word for what it does now. They change with the page (main, Disk Op., options) and become YES and NO during a question.
- **The positions (left).** The song's positions and the pattern each plays, the current one boxed: ProTracker's position editor, always in view. `BANK VIEW` shows the effect macros here instead.
- **The samples (right).** All 31 sample names, the current one boxed; empty samples dimmed. In the Disk Op. this column lists the modules instead.
- **The values (bottom).** Four boxes on each side, each above one of the 16 MUTE UP/DOWN pairs under the LCD: the four leftmost pairs and the four rightmost change the value shown right above them (section 4).



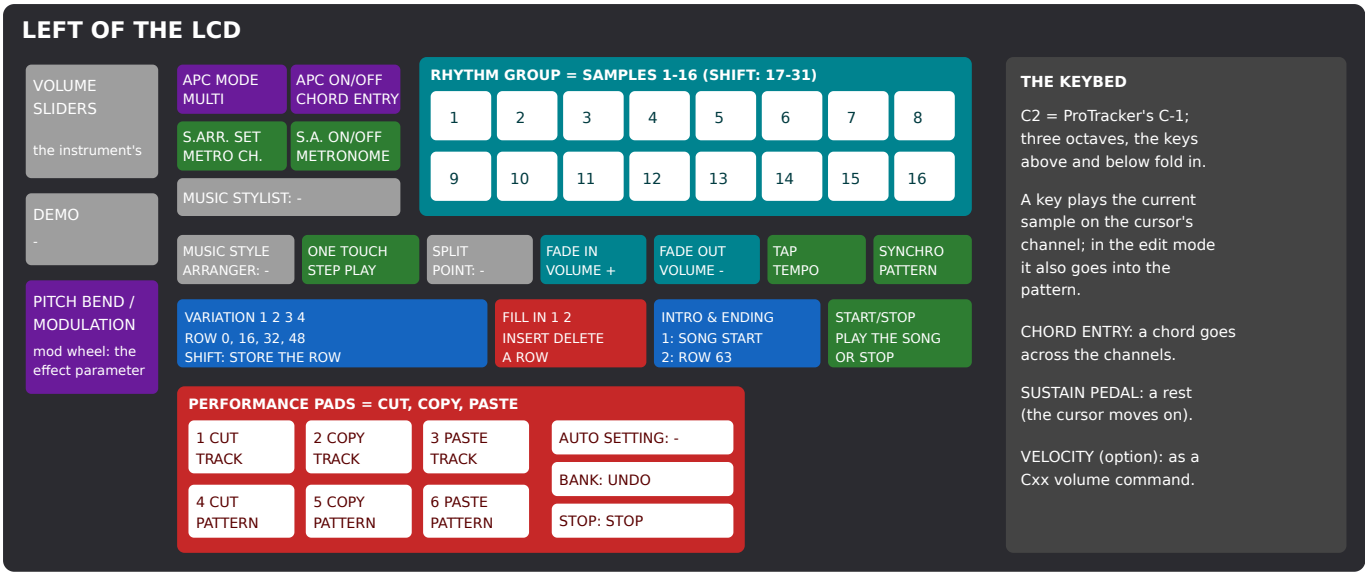
The edit mode: the red cursor on the middle row, here on channel 1's note. A chord from the keybed went into row 0 across channels 1-3 (C-2, E-2, G-2, sample 03).

ProTracker's own screen works as on the Amiga: `POS`, `PATTERN` and `LENGTH` at the top left, the current sample's number, volume, length and loop below them, the quadrascope (or spectrum analyzer) at the top right, the song's and the sample's names, the status line, the tempo (BPM), and the pattern with its VU meters. Its buttons (PLAY, DISK OP. and the rest) are drawn but not pressed: the panel does that.

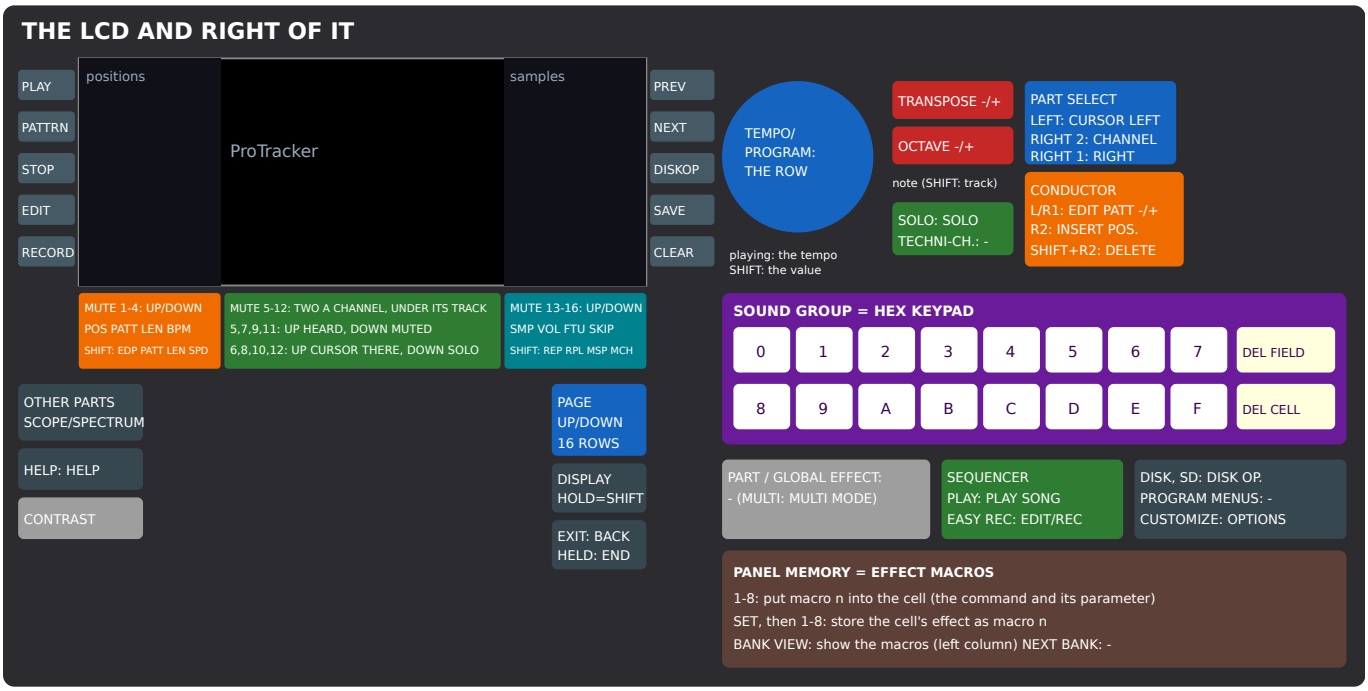
4. The panel at a glance

The panel's sections each take one of ProTracker's jobs. The two drawings show the left and right halves of the panel; the colours are the jobs:

playing · moving · entering · editing · the song · samples · effect macros · files, pages · left to the instrument



The left half of the panel. "-": does nothing in ProTracker.



The LCD, its soft keys and the MUTE pairs, and the right half of the panel.

> **SHIFT.** Hold `DISPLAY HOLD` and press another control for its second function (the TEMPO/PROGRAM wheel's LED blinks while it is held). The values at the bottom of the LCD change to show what the MUTE pairs do with SHIFT.

5. Playing modules

to	press	ProTracker's
play the song from the current position	`START/STOP`, `SEQUENCER PLAY` or `LCDL 1`	right Amiga key
play the current pattern over and over	`SYNCHRO & BREAK` or `LCDL 2`	right Alt
stop	`START/STOP`, `PADS STOP` or `LCDL 3`	space
the next or previous module	`LCDR 2`, `LCDR 1`	Disk Op.
change the tempo while playing	turn `TEMPO/PROGRAM`, or `TAP TEMPO` four times on the beat	the BPM arrows
go to another position while playing	`MUTE UP 1` / `MUTE DOWN 1` (POS)	the POS arrows
mute or hear a channel	`MUTE DOWN` / `MUTE UP` 5, 7, 9, 11 (under channels 1-4)	the mute buttons
solo a channel	`MUTE DOWN` 6, 8, 10, 12, or `SOLO` for the cursor's	right-click a mute button
scope or spectrum	`OTHER PARTS/TG`	click the visualizer

The keybed always plays: a key sounds the current sample on the cursor's channel, at its note. While a song plays, that channel's own notes take over again at the next note.

6. Editing a pattern

The edit mode

Press `EASY REC` (or `LCDL 4`). Playing stops, and the red cursor appears on the middle row. Press EASY REC again, or EXIT briefly, to leave the edit mode. While a pattern is playing, EASY REC starts the *record* mode instead (LCDL 5 too): the pattern plays over and over and what you enter goes into it at the row being played.

Moving the cursor

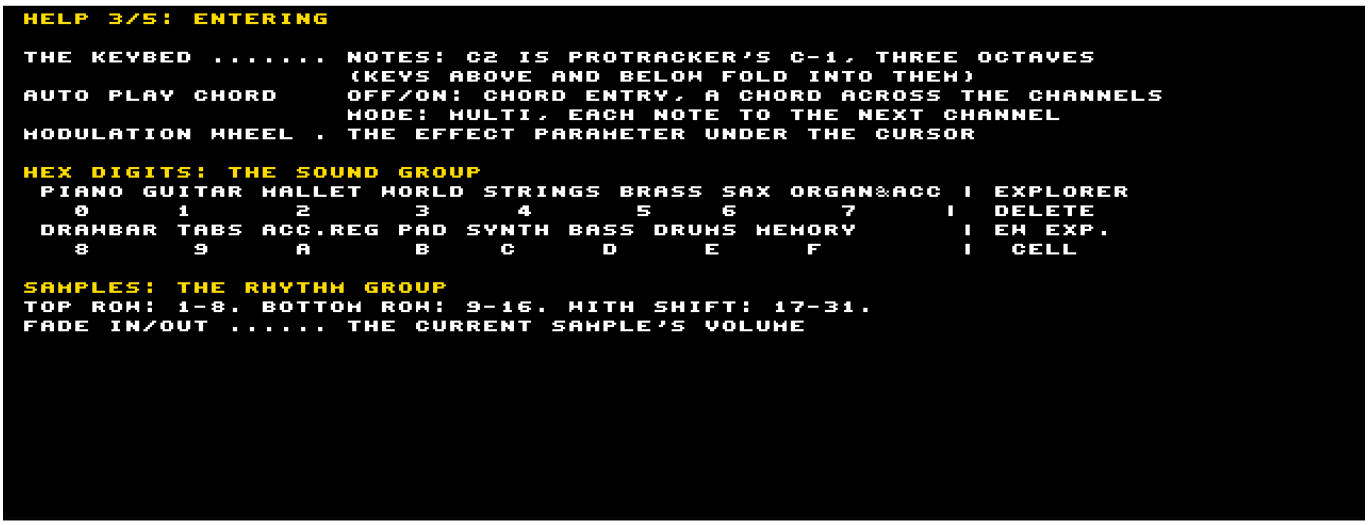
to move	press
a row up or down	turn `TEMPO/PROGRAM`
16 rows	`PAGE UP`, `PAGE DOWN`
to row 0, 16, 32, 48	`VARIATION 1`-`4` (SHIFT + VARIATION stores the current row in that button)
to row 63	`INTRO & ENDING 2`
a column left or right	`PART SELECT LEFT`, `PART SELECT RIGHT 1`

to the next channel	`PART SELECT RIGHT 2`
to channel 1-4	`MUTE UP` 6, 8, 10, 12 (the pair under that channel)

Each channel has six columns, as in ProTracker: the note, the sample number's two digits, the effect's command and its parameter's two digits.

Entering

- **Notes:** play the keybed. The note and the current sample's number go into the cell under the cursor, and the cursor moves down by the *edit skip* (shown as SKP; MUTE 16 changes it, 0 to 9).
- **Digits:** on a sample or effect column, press the SOUND GROUP button for the digit: the top row is 0-7, the bottom row 8-F. The cursor moves down by the edit skip, as ProTracker's does. Set the skip to 0 to type a whole effect in one row.
- **Chords:** turn `AUTO PLAY CHORD OFF/ON` on and play a chord: its notes go into the cursor's channel and the ones to its right, on the same row.
- **MULTI:** `AUTO PLAY CHORD MODE` or `MULTI`: each note you play goes into the next channel, so a melody can be spread over the channels as you play.
- **Rests:** the sustain pedal moves the cursor on without a note.
- **The modulation wheel** sets the effect parameter under the cursor (when the cursor is on one), 00 to FE: handy for volumes (Cxx) and slides.
- **SHIFT + TEMPO/PROGRAM** changes the value under the cursor a step at a time: the note a semitone, a digit by one.



HELP, page 3: the SOUND GROUP's digits. HELP shows five pages of these reminders on the instrument.

Changing what is there

to	press	ProTracker's
delete the note (on the note column) or the effect (elsewhere)	`SOUND EXPLORER`	Del
delete the whole cell	`EW EXPANSION` (or SHIFT + SOUND EXPLORER)	Shift+Del
insert an empty row at the cursor (the track's rows below move down)	`FILL IN 1` (SHIFT: all four tracks)	Shift+Return

delete the row at the cursor (the rows below move up)	`FILL IN 2` (SHIFT: all four tracks)	Backspace
transpose the note a semitone	`TRANSPOSE -`/`+` (SHIFT: the whole track)	Edit Op. 1
transpose the note an octave	`R1/R2 OCTAVE -`/`+` (SHIFT: the whole track)	Edit Op. 1
cut, copy, paste the track	`PAD 1`, `PAD 2`, `PAD 3`	Shift+F3, F4, F5
cut, copy, paste the pattern	`PAD 4`, `PAD 5`, `PAD 6`	Alt+F3, F4, F5
undo the last change	`PADS BANK` (again: redo)	(pt2-clone's undo)
put an effect macro into the cell	`PANEL MEMORY 1`-`8`	Alt+1-0
store the cell's effect as a macro	`SET`, then `PANEL MEMORY 1`-`8`	Ctrl+1-0
see the macros	`BANK VIEW`	
play one row and move on	`ONE TOUCH PLAY`	Return

7. Arranging the song

The song is the list of positions shown in the left column. Each line is a position and the pattern it plays.

to	press
go to another position	`MUTE UP 1` / `MUTE DOWN 1` (POS)
change the pattern this position plays	`MUTE UP 2` / `MUTE DOWN 2` (PATT)
make the song longer or shorter	`MUTE UP 3` / `MUTE DOWN 3` (LEN)
insert a position (a copy of this one, after it)	`CONDUCTOR RIGHT 2`
delete this position	SHIFT + `CONDUCTOR RIGHT 2`
edit another pattern without changing the song	`CONDUCTOR LEFT` / `CONDUCTOR RIGHT 1` (the pattern number at the screen's bottom left)
go to the song's start	`INTRO & ENDING 1`
the tempo	`MUTE UP 4` / `MUTE DOWN 4` (BPM), TAP TEMPO, or the wheel while playing

As in ProTracker, the tempo and speed are not stored in the module's header: a song sets them with the F command in its first row (for example F06 for speed 6, F7D for 125 BPM). To keep a tempo, enter it as an Fxx effect.

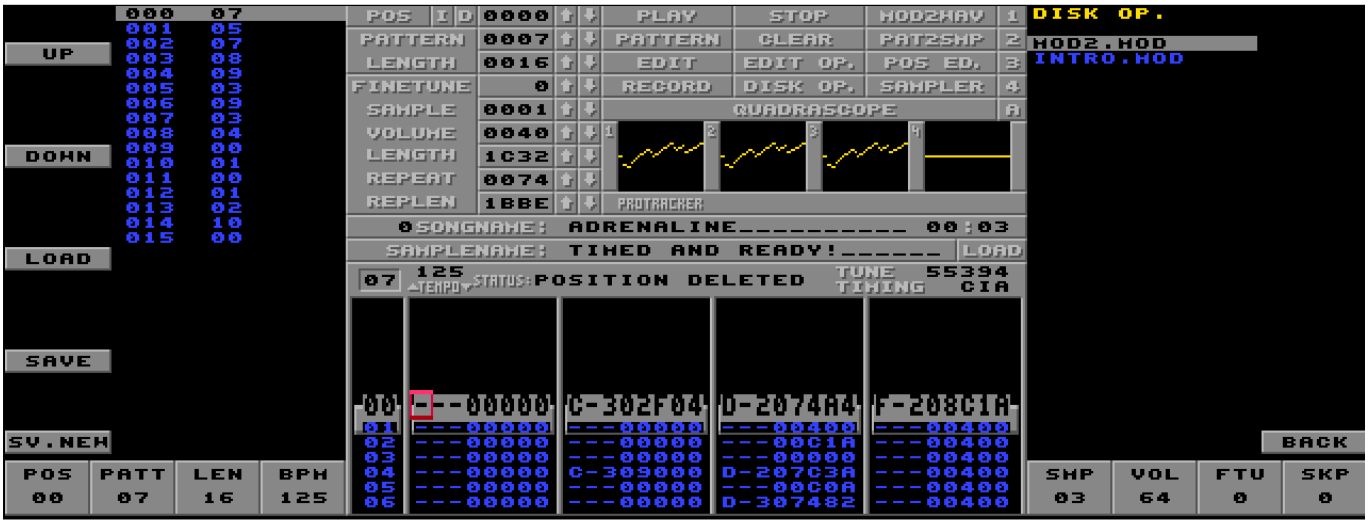
8. Samples

The current sample is the one notes play and enter. Choose it with the `RHYTHM GROUP`: its top row is samples 1-8 and its bottom row 9-16; with SHIFT, 17-31. Or step through them with `MUTE UP 13` / `MUTE DOWN 13` (SMP). The right column lists all 31.

value	pair	with SHIFT
SMP, the sample	MUTE 13	REP, the loop's start (hex, bytes)
VOL, the sample's volume (0-64); also FADE IN / FADE OUT	MUTE 14	RPL, the loop's length (hex, bytes)
FTU, its finetune (-8 to +7)	MUTE 15	MSP, the metronome's speed (rows a beat)
SKP, the edit skip (0-9)	MUTE 16	MCH, the metronome's channel (0: off)

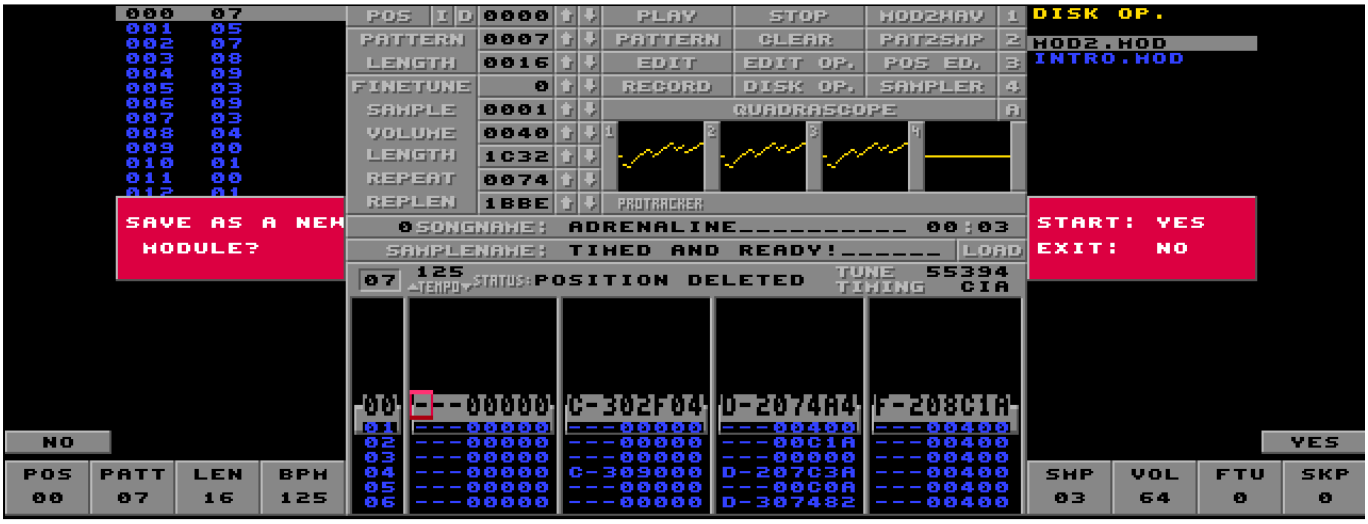
The samples' sounds themselves cannot be edited or recorded on the KN7000: there is no sample editor. The metronome (`SOUND ARRANGER OFF/ON`, and `SET` for the cursor's channel) plays sample 1F every few rows, as ProTracker's does.

9. Loading and saving



The Disk Op. (LCDR 3, DISK or SD): the modules in the right column; the soft keys change to UP, DOWN, LOAD, SAVE, SV.NEW and BACK.

- **Load:** open the Disk Op., choose with the wheel (or UP/DOWN, PAGE UP/DOWN), press LOAD. Or step with LCDR 1 / LCDR 2 from the main page. If the module in hand has unsaved edits, you are asked first.
- **Save:** LCDR 4 (or SAVE in the Disk Op.) writes the module over its own file, after asking.
- **Save as new:** SV.NEW in the Disk Op. writes `PTNEW01.MOD` (or the next free number). Your original stays as it was.
- **Clear:** LCDR 5 empties the song (its patterns and positions), after asking; the samples stay.



A question. START/STOP, SET or LCDR 5 (YES) answer yes; EXIT or LCDL 5 (NO) answer no.

Modules are saved as ProTracker 2.3D saves them ("M.K.", 31 samples), so they play in ProTracker, in every other tracker, and in pt2-clone on a computer.

10. A first session, step by step

This makes a short four-bar beat from scratch inside a module, using its samples.

1. Start ProTracker. The first module plays; press `START/STOP` to stop it.
2. Press `LCDR 3` (DISKOP), choose a module with the wheel, press `LCDL 3` (LOAD). Press START/STOP to stop it again.
3. Choose an empty pattern to work in: `CONDUCTOR RIGHT 1` until the bottom-left number shows a pattern the song does not use (the left column shows which ones it uses).
4. Press `EASY REC`: the edit mode.
5. Choose the bass drum's sample with the RHYTHM GROUP (watch the right column), then `VARIATION 1` (row 0). Set the edit skip to 4 with `MUTE UP 16`.
6. Play C3 on the keybed four times: a kick on rows 0, 4, 8 and 12. Keep going down the pattern with more notes, or with the sustain pedal for empty beats.
7. Press `PART SELECT RIGHT 2` (channel 2), choose the snare's sample, `VARIATION 1`, and enter it on the off-beats.
8. Hear it: `SYNCHRO & BREAK` plays the pattern over and over. Press it again any time; START/STOP stops.
9. An effect: back in the edit mode, set the edit skip to 0 (`MUTE DOWN 16`), move the cursor right three times to the command column, and press `SYNTH` (C), `PART SELECT RIGHT 1`, `STRINGS & VOCAL` (4), `PART SELECT RIGHT 1`, `PIANO` (0): C40, volume 40.
10. Keep it as a macro: `SET`, `PANEL MEMORY 1`. Now PANEL MEMORY 1 puts C40 into any cell.
11. Put the pattern in the song: go to the last position with MUTE 1 (POS), press `CONDUCTOR RIGHT 2` (a copy of it now follows it), MUTE UP 1 to go to that copy, and set its pattern to yours with MUTE 2 (PATT).
12. Save: `LCDR 3`, then `LCDL 5` (SV.NEW), then START/STOP. The status line says `MODULE SAVED !`.

11. Every control

control	does	with SHIFT
LCDL 1-5 (main)	PLAY, PATTERN, STOP, EDIT, RECORD	
LCDR 1-5 (main)	previous module, next module, Disk Op., SAVE, CLEAR	
MUTE 1-4	POS, PATT, LEN, BPM up/down	edited pattern, PATT, LEN, speed
MUTE 5, 7, 9, 11	channel 1-4: UP heard, DOWN muted	
MUTE 6, 8, 10, 12	channel 1-4: UP the cursor there, DOWN solo	

MUTE 13-16	SAMPLE, VOLUME, FINETUNE, edit skip up/down	REPEAT, REPLEN, metronome speed, channel
PAGE UP/DOWN	16 rows (Disk Op.: 16 modules; HELP: pages)	
DISPLAY HOLD	SHIFT	
EXIT	briefly: back (page, macros, edit mode); held 1.5 s: the end	
HELP	five pages of help (PAGE UP/DOWN, the wheel); again or EXIT: back	
OTHER PARTS/TG	quadrascope or spectrum analyzer	
TEMPO/PROGRAM	the row; while playing, the tempo; in the Disk Op., the module	the value under the cursor
TRANSPOSE -/+	the note a semitone	the track a semitone
R1/R2 OCTAVE -/+	the note an octave	the track an octave
PART SELECT	LEFT: cursor left; RIGHT 1: cursor right; RIGHT 2: next channel	
CONDUCTOR	LEFT/RIGHT 1: the edited pattern; RIGHT 2: insert a position	RIGHT 2: delete the position
SOLO	solo the cursor's channel (again: all heard)	
SOUND GROUP	hex digits 0-7 (top), 8-F (bottom)	
SOUND EXPLORER	delete the note or the effect	delete the cell
EW EXPANSION	delete the cell	
RHYTHM GROUP	samples 1-16	samples 17-31
AUTO PLAY CHORD	OFF/ON: chord entry; MODE: MULTI	
MULTI	MULTI	
SOUND ARRANGER	OFF/ON: metronome; SET: on the cursor's channel	
ONE TOUCH PLAY	play one row and move on	
FADE IN / OUT	the sample's volume up / down	
TAP TEMPO	the tempo from taps, a tap a beat	
SYNCHRO & BREAK	play the pattern over and over	
VARIATION 1-4	row 0, 16, 32, 48	store the current row there
FILL IN 1 / 2	insert / delete a row in the track	in the pattern
INTRO & ENDING 1 / 2	the song's start / row 63	

START/STOP	play the song, or stop; a question: yes	
PADS 1-6	cut, copy, paste the track (1-3) and the pattern (4-6)	
PADS BANK / STOP	undo / stop	
SEQUENCER PLAY / EASY REC	play the song / the edit mode (playing: record)	
DISK, SD	the Disk Op.	
CUSTOMIZE	options: blank zeroes, dots, sharps or flats, velocity as Cxx, the scope	
PANEL MEMORY 1-8	the effect macro into the cell (after SET: store)	
SET	store a macro (then 1-8); a question: yes	
BANK VIEW	show the macros	
keybed	notes (C2 = C-1); chords with chord entry	
sustain pedal	a rest	
modulation wheel	the effect parameter under the cursor	
left to the instrument	the volume sliders, CONTRAST; nothing: DEMO, MUSIC STYLIST, MUSIC STYLE ARRANGER, SPLIT POINT, TECHNI-CHORD, PROGRAM MENUS, CUSTOM PANEL, FAVORITES, NEXT BANK, PADS AUTO SETTING, the PART and GLOBAL EFFECT buttons (but MULTI), the pitch-bend wheel	

While ProTracker runs, the instrument's own functions behind these buttons are kept from the instrument: pressing PANEL MEMORY 1 puts an effect into a cell and does not recall a registration.

12. ProTracker's effects

An effect is a command digit and a two-digit parameter (x and y are its digits, xx both). Effects work on the channel they are in, at that row.

effect	what it does
`0xy`	arpeggio: the note, +x, +y semitones in turn
`1xx`	slide the pitch up
`2xx`	slide the pitch down
`3xx`	slide to the note (portamento)
`4xy`	vibrato: speed x, depth y
`5xy`	3 (go on) and volume slide
`6xy`	4 (go on) and volume slide
`7xy`	tremolo: speed x, depth y
`9xx`	start the sample xx00 bytes in
`Axy`	volume slide: up x or down y
`Bxx`	jump to position xx
`Cxx`	volume xx (00-40)
`Dxy`	to the next position, row xy (decimal)
`Fxx`	speed (01-1F) or tempo in BPM (20-FF)

effect	what it does
`E0x`	the Amiga's filter (not heard here)
`E1x`	fine slide up
`E2x`	fine slide down
`E3x`	glissando on/off
`E4x`	vibrato waveform
`E5x`	finetune
`E6x`	loop: E60 marks, E6x plays x times
`E7x`	tremolo waveform
`E9x`	retrigger every x ticks
`EAx`	fine volume up
`EBx`	fine volume down
`ECx`	cut the note after x ticks
`EDx`	delay the note x ticks
`EEx`	delay the pattern x rows
`EFx`	invert loop (not heard here)

13. What differs from the Amiga

- The pattern shows 13 rows (the LCD has 240 lines); the VU meters are a little shorter.
- The sound is mono and unfiltered: the Amiga's left/right channels and its filters (E0x) are not there.
- A sample loop that starts on an odd word may begin up to three samples early, once.
- A loop changed while a sample plays takes over up to a tick late.
- EFx changes only ProTracker's copy of the sample, not what the DSP plays.
- There is no sample editor, no sampling, no MOD2WAV, and packed (PowerPacker, XPK) modules do not load.
- Modules are listed by their 8.3 names, not their long names.

14. If something goes wrong

what you see	what to do
`CAN'T FIND FILE !`	the module could not be read from the card: check that the card is in, and that the file is in PROTRACK\.
`NOT A MOD FILE !`, `UNSUPPORTED MOD !`	not a 4-channel module, or a packed one.
`NO DSP: SILENT !`, `DSP NOT READY !`	the effects DSP could not be used: the program runs silent. Leave and start again.
`NOT IN EDIT MODE`	press EASY REC first.
`MODULE TOO BIG !`	the module does not fit in memory for saving (more than 376 KB).
`LIST IS FULL !`	PROTRACK\ has 128 modules, or PTNEW99.MOD exists: move some off the card.
a button does nothing	check the page (the tags) and whether SHIFT is held; HELP lists everything.

Modules up to 376 KB load, with up to 268 KB of samples.